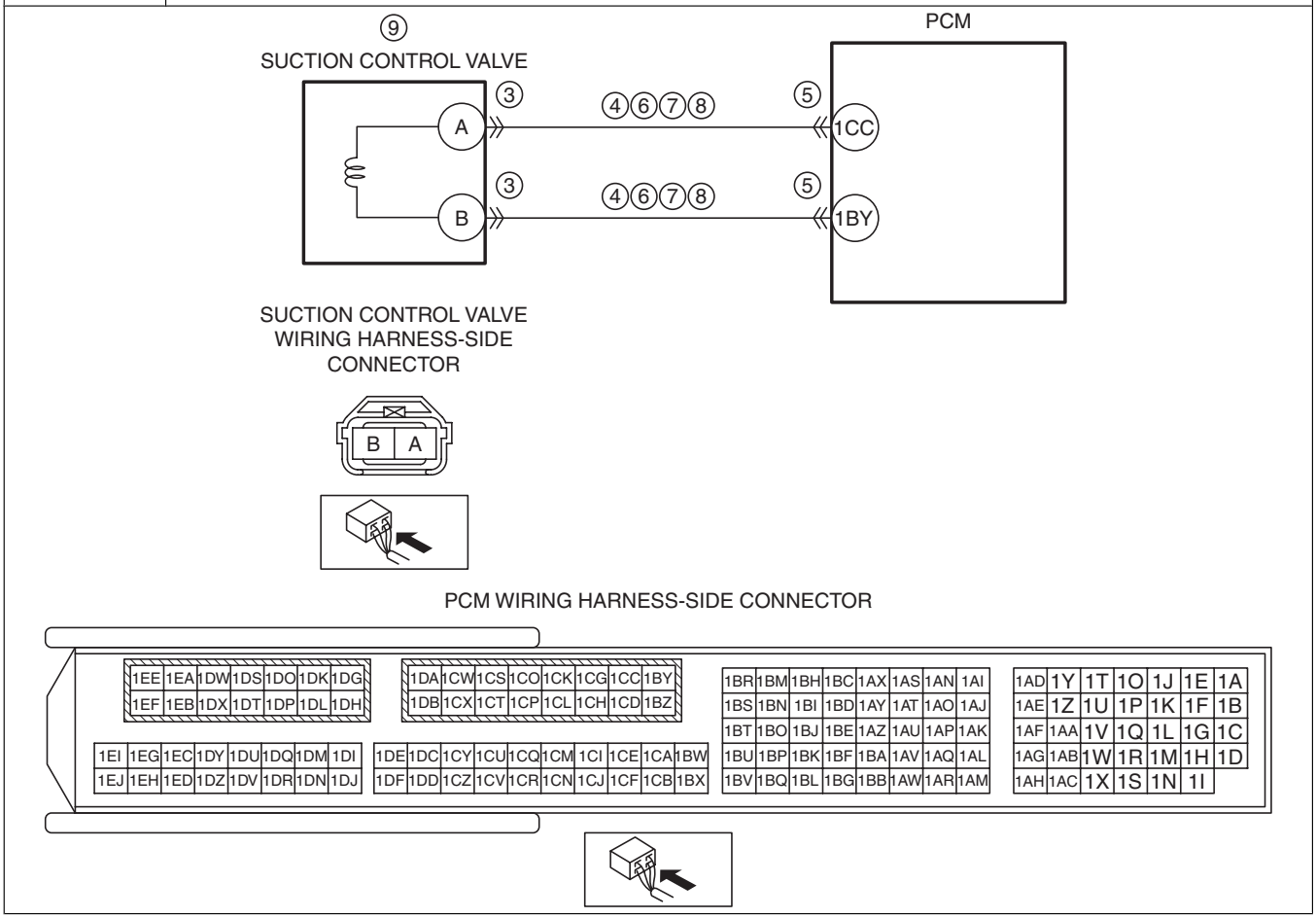
DTC P062A:00 [SKYACTIV-D 1.5]

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DTC P062A:00	Suction control valve circuit problem
DETECTION CONDITION	• Any of following conditions occurs: — With the following conditions met, a condition has continued for **1 s** in which the suction control valve control current is **1.0 A or less** or the control voltage is more than the specified value. **MONITORING CONDITIONS** • Battery voltage: **8 V or more** • **0.5 s** have elapsed after the ignition was switched ON (engine on) — With the following conditions met, a condition has continued for **1 s** in which the suction control valve control voltage is the specified value or less or the control current is **more than 1.0 A**. **MONITORING CONDITIONS** • Battery voltage: **8 V or more** • **0.5 s** have elapsed after the ignition was switched ON (engine on) **Diagnostic support note** • This is an intermittent monitor (fuel system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Limits the engine torque or the upper limit of the engine speed. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Suction control valve connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Suction control valve terminal A—PCM terminal 1CC — Suction control valve terminal B—PCM terminal 1BY • PCM connector or terminals malfunction • Short to power supply in wiring harness between the following terminals: — Suction control valve terminal A—PCM terminal 1CC — Suction control valve terminal B—PCM terminal 1BY • Suction control valve circuits are shorted to each other • Open circuit in wiring harness between the following terminals: — Suction control valve terminal A—PCM terminal 1CC — Suction control valve terminal B—PCM terminal 1BY • Suction control valve malfunction • PCM malfunction

DTC P062A: 00	Suction control valve circuit problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (fuel system related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT SUCTION CONTROL VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the suction control valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT SUCTION CONTROL VALVE CIRCUIT FOR SHORT TO GROUND** • Verify that the suction control valve connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Suction control valve terminal A — Suction control valve terminal B • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Suction control valve terminal A—PCM terminal 1CC — Suction control valve terminal B—PCM terminal 1BY **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 10.
		No	Go to the next step.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
6	**INSPECT SUCTION CONTROL VALVE CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the suction control valve and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the following terminals (wiring harness-side): — Suction control valve terminal A — Suction control valve terminal B • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Suction control valve terminal A—PCM terminal 1CC • Suction control valve terminal B—PCM terminal 1BY **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 10.
7	**INSPECT SUCTION CONTROL VALVE CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the suction control valve and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between suction control valve terminals A and B (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Suction control valve terminal A—PCM terminal 1CC • Suction control valve terminal B—PCM terminal 1BY **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 10.
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**INSPECT SUCTION CONTROL VALVE CIRCUIT FOR OPEN CIRCUIT** • Verify that the suction control valve and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Suction control valve terminal A—PCM terminal 1CC — Suction control valve terminal B—PCM terminal 1BY • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Suction control valve terminal A—PCM terminal 1CC • Suction control valve terminal B—PCM terminal 1BY **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
9	**INSPECT SUCTION CONTROL VALVE** • Inspect the suction control valve. (See SUCTION CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the suction control valve, then go to the next step. (See SUCTION CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.